“stacking” risks, the expected return of any asset class can be estimated. This section takes you through the process of stacking risk premiums to derive an expected return from an asset class.

Calculating an expected return for any asset class starts with T-bill risk and return because that is the safest investment you can own. The T-bill return can be divided into two parts: an expected inflation component and a real risk-free return component.

T-bill yields = expected inflation over the maturity period + real risk-free rate

Since Treasury bills are the safest investment available, all other investments must earn at least the T-bill’s return. In addition, all other investments must have an extra return to compensate for the risks inherent in the particular investment. The extra return is known as a *risk premium.*

Asset class expected investment return = T-bill yield + risk premium

Every investment has an expected risk premium above and beyond T-bills that is based on that investment’s unique risks. For example, 10-year Treasury bonds have term risk. That is the price risk that results from interest-rate changes that may occur after you buy a 10-year bond through its maturity date. If interest rates rise during that period, you lose out on the higher rates because your money will have been tied up in the 10-year note. Bonds with longer maturities have even greater term risk, and the greater the term risk, the higher the expected return of the bond. This is why long-term bonds yield more than short-term bonds.

Duration is an approximation of term risk in a bond. Basically, it is an estimate of price movement given a 1 percent move in general interest rates. Duration is calculated based on a bond’s maturity date and its stated interest rate. Assume that a 10-year Treasury bond has a 7-year duration. This approximates a 7 percent loss in market value if interest rates go up by 1 percent.

Figure 11-5 highlights the term risk derived from Treasury Inflation-Protected Securities (TIPS) based on different maturities and durations. Since the inflation rate is already factored out of TIPS yields, the curve shown in Figure 11-5 represents pure term risk. See Chapter 8 for more information on TIPS.